1. Reconstruct the line plot featuring 17th century births in England shown in the *A Grammar of Graphics* textbook chapter using `ggplot2` code. The data frame that you will need to access is called `arbuthnot`; both are located in the `stat20data` package. Write the code you used below for each one, as well as the code that you used in any steps that you might need to take before making the visualization.

2. Now, access the `penguins` data frame. Consider a new metric called `bill_size` that's the sum of the length and depth. What is the average bill size and its standard deviation among each species, broken out among each of the island? Sort your resulting data structure in decreasing order by average bill size. *Your answer to this question should consist of one `dplyr` data pipeline.*

3. There is a data set built into R called `mtcars` that includes several measures on different types of cars. Learn more about the data set using `?mtcars`. What is the better way to compare the strength of the linear relationship between these two pairs of variables (mpg and wt; mpg and hp): the correlation coefficients or the slopes of the linear models? Why? Explain in 1-2 sentences.

4. Fit the multiple linear regression model from the *Multiple Linear Regression* slides (you may view the previous semester's slides), but instead having `"Chinstrap"` as the reference level in the `species` variable. Write the code you used here. Then, interpret the coefficient corresponding to the `"Adelie"` dummy variable in the context of the problem in 1-2 sentences.